

AMOGH UKKADGATRI

Electronics & Communication Engineer | Embedded Systems & Signal Processing
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SUMMARY

Electronics and Communication engineer with hands-on R&D experience in acoustic sensing, embedded systems, signal processing, and radar at a defence technology company. Skilled in signal acquisition, ESP32 and Raspberry Pi firmware, and source-localization methods, with proven ability to own hardware end to end, from soldering and firmware to data extraction and custom test infrastructure. Suited to regulated, high-reliability work across defence and aerospace.

EXPERIENCE

R&D Intern — SSS Defence

Sep 2025 – Present

Bengaluru, India (On-site) | DeepTech R&D, Acoustic C-UAS & Radar

- Recruited into an acoustic counter-UAS program on the strength of a final-year sound source localization project; joined the hardware team and contributed across core signal-acquisition domains.
- Part of R&D team on the microphone arrays for the first C-UAS prototype on Raspberry Pi 4, working across multiple communication protocols and gaining deep familiarity with the platform.
- Adapted a triangulation-based localization approach to support the team's advanced & classical physics-based algorithms, improving the C-UAS localization pipeline.
- Owned the full electronics build for the second prototype: soldering, ESP32 firmware, flashing, and data extraction on an independent module.
- Built a complete test rig single-handedly to validate custom mic arrays using ESP32 I2S for PDM capture; solved the classic ESP32's single-RX-channel limit with TDM to read multiple PDM mics.
- Applied networking fundamentals across prototypes: DHCP, static IP assignment, and subnet masking.
- Now contributing to a new low-power X-band radar project, building working knowledge of the radar physics, electronics, and full system operation.

PROJECTS

3D Sound Source Localization using Mic Array

Sep 2025 – March 2026

- Building a real-time audio sensing system that identifies and localizes sound sources using Classical methods and a microphone array.

Real-time 3D Radar Tracking & Classification System

Sep 2024 – Nov 2024

- Designed a real-time radar system to track objects in three dimensions and classify them by their distinct signatures.

Automatic Rain Detection System

Mar 2024 – May 2024

- Built a smart rain-detection module using embedded components for automation.

Battery Level Indicator (Analog Circuits)

Dec 2023 – Feb 2024

- Analog voltage-divider based battery monitoring system.

Centroid Finder in Python

Sep 2023 – Oct 2023

- Python script that computes centroid location from coordinate inputs.

Smart Alcohol Detection & Engine Immobilization System

Jan 2023 – Mar 2023

- Created a breath-alcohol sensor integrated with a vehicle ignition lock to immobilize the engine on detection.

SKILLS

Languages & Tools: C, Python, Verilog (basics), MATLAB, Audacity, Linux

Hardware & Embedded: ESP32, 8051, Raspberry Pi, Arduino, I2S/PDM/TDM audio, sensors, op-amps, MOSFETs, transistors, soldering & firmware flashing

Networking & Domains: DHCP, static IP, subnet masking, Ethernet data transfer; embedded systems, signal processing, source localization, radar fundamentals

Soft Skills: Teamwork, leadership, time management, communication, problem solving

EDUCATION

B.E., Electronics & Communication — KLE Institute of Technology

2022 – 2026

CGPA: 8.84 / 10

Senior Secondary (XII), Science — Chetan PU College, Karnataka State Board

2022 | 81.00%

Secondary (X) — Oriental Public School, State Board

2020 | 96.16%

POSITIONS OF RESPONSIBILITY & ACTIVITIES

IEEE: Chair, KLEIT-IEEE MTT-S Chapter (2024–25) · Vice-Chair, KLEIT-IEEE Student Branch (2023–24)

College: Coordinator, Technical & Cultural Fest · Class Representative (2023–24) · University Cultural Group (Theatre)